MILITARY SPECIFICATION

SCALES, GRAPHICAL FIRING: FAN TYPE,

155-MM HOWITZER, FT-155-Q-3 and 105-MM HOWITZER, FT-105-H-6

1. SCOPE

1.1 This specification covers fan-type graphical firing scales used for determining and plotting data pertinent to indirect firing of artillery weapons (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Ordnance Materiel, Packaging of
MIL-I-45208	Inspection System Requirements
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1220

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

DRAWINGS

U. S. Army Munitions Command

F8287537	Scale, Graphical Firing: Fan Type, 155-MM Howitzer, FT-155-Q-3
F8287545	Scale, Graphical Firing: Fan Type, 105-MM Howitzer, FT-105-H-6

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The scale specified in the contract or order shall be manufactured in accordance with the applicable drawing listed in Section 2 and all drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Part identification and marking
(d) Workmanship

3.3 Environmental.

3.3.1 Operating temperatures.- The fan scale, ballistic scales, and cursor shall assemble freely, and the cursor shall operate smoothly with the scales, while exposed to, and thermally stabilized at, ambient temperatures of -40° and +125°F.

3.3.2 Storage temperatures.- The complete graphical firing scale shall meet all performance requirements of this specification at a standard ambient temperature of between 60° and 90°F after having been exposed to, and thermally stabilized at, ambient temperatures of -60° and +140°F. There shall be no color changes, separations, bubbles, tears, or cracks, and no warpage in excess of 0.062 inch as measured at the outer edges of the fan and ballistic scales.

3.4 Printing and surface quality.- Graduation lines, letters, numerals, and triangles on the various scales and the lines on the cursor shall be applied in black or red as indicated on pertinent drawings. They shall be sharp, evenly colored, and without skips or breaks. Letters and numerals shall not touch each other or any line of any scale, with the exception of the range and 100/R scales on the ballistic scales.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

On these, some graduation lines will intersect the identifying numerals as shown on pertinent drawings. Surfaces of scales shall be free from stains and other imperfections.

3.5 Cursor.- The top frosted surface of the cursor shall be capable of accepting clear and legible markings from a lead pencil of 2H hardness. Upon erasure with a commercial pencil eraser, there shall be no evidence of permanent marks, smudges, or smears.

3.6 Fan scale.- The frosted area of the fan scale's top surface shall be capable of meeting the requirements specified in 3.5 for the cursor. The fan scale shall permit attachment and detachment of the ballistic scales.

3.7 Accuracy.

3.7.1 Ballistic scales.- With each ballistic scale assembled separately on the fan, and with the cursor assembled to slide on the ballistic scale, readings indicated in Tables I through VI shall be obtained, within the tolerances specified, for each range listed.

3.7.2 Mil scale.- The fan's mil scale graduations representing 400, 800, and 1200 mils, measured from the vertex and the left edge of the fan's range scale, shall be accurately positioned within the thickness of a graduation line as determined at the perimeter of the mil scale.

3.7.3 Range scales.- Range scale graduations on the ballistic scales shall coincide with their counterpart graduations on the range scale of the fan within 0.015 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

TABLE I

SCALE, GRAPHICAL FIRING:FAN TYPE,
105-MM HOWITZER, FT-105-H-6
USING BALLISTIC SCALE-8287546

LOW ANGLE FIRING DATA

SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 2	3,000	410.5 ± 2.5	17.02 ± 0.10
Charge 7	5,000	185.1 ± 0.7	15.68 ± 0.05
	10,000	547.7 ± 1.6	39.74 ± 0.10

HIGH ANGLE FIRING DATA

SCALE	RANGE (Meters)	ELEVATION (Mils)	DRIFT (Mils)	TF (Seconds)	10 m SI (Mils)
Charge 1	3,000	1040.0 ± 4.0	43.4 ± 0.6	32.75 ± 0.09	5.92 ± 0.17
Charge 2	3,500	1032.8 ± 3.6	42.45 ± 0.50	35.09 ± 0.07	6.19 ± 0.20
Charge 3	4,200	1019.5 ± 3.3	40.76 ± 0.40	38.03 ± 0.07	6.81 ± 0.17
Charge 4	5,200	1008.1 ± 2.8	39.39 ± 0.33	42.12 ± 0.07	7.42 ± 0.19
Charge 5	6,500	1026.5 ± 2.1	41.65 ± 0.26	48.35 ± 0.06	6.09 ± 0.11
Charge 6	8,000	1041.7 ± 1.6	43.61 ± 0.21	55.02 ± 0.05	5.27 ± 0.07
Charge 7	10,000	1013.2 ± 1.4	39.98 ± 0.17	60.44 ± 0.06	6.22 ± 0.08

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

TABLE II

SCALE, GRAPHICAL FIRING; FAN TYPE;
105-MM HOWITZER, FT-105-H-6
USING BALLISTIC SCALE-8287547

LOW ANGLE FIRING DATA			
SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 1	2,500	393.9 ± 2.8	15.13 ± 0.10
Charge 5	3,000 6,000	184.5 ± 0.9 451.1 ± 1.6	11.34 ± 0.06 26.07 ± 0.09
Charge 6	4,000 7,000	196.4 ± 0.8 411.7 ± 1.2	13.81 ± 0.05 27.20 ± 0.07

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

TABLE III

SCALE, GRAPHICAL FIRING: FAN TYPE
105-MM HOWITZER, FT-105-H-6
USING BALLISTIC SCALE-8287548

LOW ANGLE FIRING DATA			
SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 3	2,000	200.3 ± 1.5	9.51 ± 0.07
	4,000	488.9 ± 2.7	21.91 ± 0.11
Charge 4	2,500	200.8 ± 1.2	10.71 ± 0.06
	4,500	421.9 ± 1.9	21.49 ± 0.09

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

TABLE IV

SCALE, GRAPHICAL FIRING: FAN TYPE,
155-MM HOWITZER, FT-155-Q-3
USING BALLISTIC SCALE 8287539

LOW ANGLE FIRING DATA

SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 2	2,000	196.0 ± 1.4	9.29 ± 0.07
	4,000	467.0 ± 2.4	21.06 ± 0.10
Charge 7	7,000	180.7 ± 0.6	18.37 ± 0.05
	13,000	521.8 ± 1.2	44.08 ± 0.08

HIGH ANGLE FIRING DATA

SCALE	RANGE (Meters)	ELEVATION (Mils)	DRIFT (Mils)	TF (Seconds)	10 # SI (Mils)
Charge 1	3,500	1020.7 ± 4.0	36.44 ± 0.32	34.70 ± 0.09	7.18 ± 0.25
Charge 2	4,500	991.0 ± 4.0	33.71 ± 0.26	38.08 ± 0.09	9.88 ± 0.36
Charge 3	5,500	1026.7 ± 2.6	36.4 ± 0.2	44.02 ± 0.06	6.84 ± 0.15
Charge 4	7,000	1027.8 ± 2.0	36.48 ± 0.16	50.16 ± 0.06	6.54 ± 0.11
Charge 5	8,500	1062.6 ± 1.4	39.42 ± 0.13	57.61 ± 0.05	5.03 ± 0.06
Charge 6	11,000	1024.2 ± 1.4	36.21 ± 0.11	63.68 ± 0.06	6.82 ± 0.09
Charge 7	13,000	1066.6 ± 1.0	39.77 ± 0.09	73.51 ± 0.05	4.62 ± 0.04

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

TABLE V

SCALE, GRAPHICAL FIRING; FAN TYPE;
155-MM HOWITZER, FT-155-Q-3
USING BALLISTIC SCALE-8287540

LOW ANGLE FIRING DATA			
SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 1	3,000	426.3 ± 2.6	17.19 ± 0.10
Charge 5	4,000	183.7 ± 0.7	13.00 ± 0.05
	8,000	462.5 ± 1.3	30.48 ± 0.08
Charge 6	5,000	168.2 ± 0.6	14.44 ± 0.05
	10,000	467.3 ± 1.1	35.20 ± 0.07

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

TABLE VI

SCALE, GRAPHICAL FIRING: FAN TYPE;
155-MM HOWITZER, FT-155-Q-3
USING BALLISTIC SCALE-8287541

LOW ANGLE FIRING DATA

SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 3	3,000	233.6 ± 1.2	12.53 ± 0.07
	5,000	456.0 ± 1.9	23.41 ± 0.09
Charge 4	3,500	204.5 ± 0.9	12.73 ± 0.06
	6,500	464.5 ± 1.6	27.31 ± 0.09

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Table VII titled "Classification of Defects". Examinations and tests for packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

TABLE VII - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

CRITICAL: NONE		REQUIREMENT	TEST PROCEDURE
MAJOR: AQL 1.0% Defective			
101.	Printing and surface quality	3.4	4.5.4
102.	Cursor	3.5	4.5.5
103.	Fan scale	3.6	4.5.6
104.	Ballistic scale accuracy	3.7.1	4.5.7
105.	Mil scale accuracy	3.7.2	4.5.8
106.	Range scale accuracy	3.7.3	4.5.9
MINOR: None defined			

NOTE: The tests of Table VII shall be conducted at standard ambient temperature (+60° to +90°F).

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests.

4.3.1 General.- One graphical firing scale of each model being procured shall be selected at random from each 150 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Fabrication	3.1	4.5.1, per drawing
General specification	3.2	Visual, MIL-F-13926

4.3.2 Environmental.- Three graphical firing scales of each model being procured shall be selected at random from each 100 procured, or from each month's production, whichever occurs first. The control samples shall be subjected to the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Operating temperatures	3.3.1	4.5.2
Storage temperatures	3.3.2	4.5.3

4.3.3 Control test failure.- Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL-I-45208, and all applicable requirements specified in Specification MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Measuring and Test Equipment" and "Government-Furnished Material" requirements of Specification MIL-I-45208 and all applicable requirements specified in Specification MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 Fabrication.- The graphical firing scale specified in the contract or purchase order shall be subjected to a visual, tactile, and dimensional inspection for conformance to the applicable drawings. Standard measuring equipment shall be used in performing dimensional inspection. The graphical firing scale shall conform to the requirements of 3.1.

4.5.2 Operating temperatures.- The test shall be in accordance with the "Test Facilities" requirements of Specification MIL-F-13926 and the conditions of 3.3.1. While thermally stabilized at -40° and +125°F, respectively, the assembly and operability of the fan scale, ballistic scales, and cursors shall be conducted by means of a visual and tactile examination. The components of the graphical firing scale shall conform to the requirements of 3.3.1.

4.5.3 Storage temperatures.- Test equipment shall be in accordance with the "Test Facilities" requirements of Specification MIL-F-13926 and the conditions of 3.3.2. During temperature cycling, the fan scale, ballistic scales and cursors (that comprise a graphical firing scale) shall lay on a flat surface in the test chamber. The flat surface shall be capable of supporting the entire length and width of each component. Upon return to standard ambient temperature, the fan scale, ballistic scales, and cursors shall be subjected to a visual examination for evidence of color change, separations, bubbles, tears, and cracks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

The fan and ballistic scales shall be placed flat on an appropriate surface plate and measured at the outer edges with a thickness or feeler gage for evidence of warpage. The graphical firing scale shall conform to the requirements of 3.3.2. The graphical firing scale shall then be subjected to the tests in 4.5.4 through 4.5.9, inclusive, and shall conform to the requirements specified.

4.5.4 Printing and surface quality.- The printing and surface quality of the fan scale, ballistic scales, and cursor shall be visually inspected with the aid of a magnifier of at least 3 power (3X). The printing and surface quality of the scales shall conform to the requirements of 3.4.

4.5.5 Cursor.- Perform this test with a commercial type quality lead pencil of 2H hardness. The pencil shall be equipped with a conical writing point and a clean commercial eraser. Lead pencil markings in the form of numerals and letters shall be placed throughout the frosted area of the cursor. All numerals and letters shall be clear, legible and without evidence of "skip". The frosted area of the cursor shall conform to the requirements of 3.5 when cleaned with the pencil eraser.

4.5.6 Fan scale.- The frosted area of the fan scale shall be tested in accordance with 4.5.5 and shall conform to the requirements of 3.6. The attachment and detachment of each ballistic scale to the fan scale shall be tested by means of a tactile examination. The fan scale shall conform to the requirements of 3.6.

4.5.7 Ballistic scale accuracy.- Each ballistic scale of the graphical firing scale shall be subjected to a tactile-visual examination to determine accuracy of its firing data. The firing data accuracy (low and high angle) of the ballistic scale shall be checked against the applicable table of 3.7. Superimpose the cursor index line on the given range line, and record the firing data values (EL, FS, DFT, TF, and 10M SI, as applicable) obtained at the intersection of the cursor index line with the applicable scales for the given charge. The accuracy of the ballistic scales shall conform to the requirements of 3.7.1.

4.5.8 Mil scale accuracy.- The accuracy of the mil scale graduations on the fan scale shall be verified by comparison to a calibrated template. The mil scale graduations shall conform to the requirements of 3.7.2.

4.5.9 Range scale accuracy.- Position each ballistic scale, in turn, on the fan scale and measure the alinement of the range graduations of the ballistic scale with those of the fan scale. Alinement of the range scale graduations of the ballistic scales with those on the fan scale shall be measured with an appropriate optical gage. The optical gage shall be capable of measuring any misalinement of the scale graduations within the tolerance specified. The range scale graduations of the ballistic scales shall conform to the requirements of 3.7.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Specification MIL-P-14232 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The complete graphical firing scale consists of a fan-type scale, three ballistic scales, and a cursor with hairline. It is used for determining firing data and for measuring angles, ranges, and distances in conjunction with charts or maps associated with indirect firing of artillery weapons. Scale 8287537 is used with 155-mm howitzers to which firing table FT-155-Q-3 is applicable. Scale 8287545 is utilized with 105-mm howitzers to which firing table FT-105-H-6 is pertinent.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable Packaging Data Sheet number.
(c) Selection of applicable levels of preservation, packaging, and packing.
(d) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A157Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:10Z
Check the source to verify that this is the current version before use.